---Intrusion Detection System

Using Motion Sensor

A Capstone Project

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College of Information Technology Education

PHINMA University of Iloilo

In Partial Fulfillment

of the Requirements for the degree

Bachelor of Science in Information Technology

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Approval Sheet

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ACKNOWLEDGEMENT

First and foremost, we express our deepest gratitude to Almighty God for His endless blessings, wisdom, and strength, which guided us throughout this project. Without His grace, this work would not have been possible.

We would also like to extend my sincere appreciation to our adviser, Mr. Ryan C. Valeriano, for his expert guidance, insightful feedback, and continuous encouragement. His support was crucial in shaping the direction and success of this work.

We deeply appreciate the unwavering support of my family and friends, whose encouragement and motivation kept me going during challenging times.

Finally, we acknowledge the efforts of each and everyone, especially to our group mates for their dedication, hard work, and collaboration in making this project a success.

Thank you all, and may God bless you abundantly!

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ABSTRACT

Security of our homes is an important issue for many individuals in today's world. This research focuses on developing an Intrusion Detection System using Motion Sensors. The system uses motion sensors to identify any unusual activities within a residence and sends immediate notifications to the homeowner through the mobile application. This enables homeowners to stay informed about potential intrusions and respond, even when they are away from their residence.

This project aims to establish a straightforward, dependable, and efficient security system that minimizes false alarms while functioning seamlessly with smart home technologies. The mobile application is crafted to be intuitive, offering real-time notification to keep the homeowner updated about their property's security status.

The system was tested in different scenarios to check its accuracy, reliability, and ease of use. Results show that the system is effective in improving home security by providing quick and convenient alerts. This project offers an affordable and practical way to protect homes and helps homeowners feel safer and more in control.

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CHAPTER I

Introduction of the Study

In today’s world, security to our home has become a priority for many households. Traditional security measures like locks or simple alarm systems are no longer enough to protect homes from unauthorized access. With the rise of smart technologies, modern solutions are needed to make homes safer and more secure.

This capstone project, Intrusion Detection System for Houses, aims to develop a smart security system that detects unauthorized movements in a house using motion sensors. The system will send instant notifications to the homeowner’s whenever unusual movement is detected, allowing them to take immediate action, even if they are away from home.

The goal of this project is to design a reliable and user-friendly security system that integrates motion sensors with mobile technology. By providing real-time alerts and ensuring ease of use, this study hopes to create an efficient way to enhance home security and give homeowners peace of mind.

Technical Background of the Study

Home security is an important concern for many households, especially with the increasing number of break-in incidents and unauthorized entry attempts. Traditionally, homeowners relied on physical locks or basic alarms, which often provide limited protection and no real-time awareness. These traditional methods may not alert the homeowner instantly, especially when no one is inside the house. Because of this, modern solutions are now needed that can provide immediate notifications and real-time monitoring even from remote locations.

An Intrusion Detection System using Motion Sensor is designed to improve home security by detecting motion and notifying users. This type of system integrates several modern technologies to create a reliable and automated security solution. Some of the key technologies used in this study include:

1. **PIR (Passive Infrared) Motion Sensing**: PIR sensors detect changes in infrared energy caused by movement of people within a certain area. These sensors are affordable, energy-efficient, and reliable for detecting unauthorized motion inside a house.

2. **Microcontroller(ESP32 Module)**: The microcontroller processes the signals captured by the PIR sensor. It communicates with the server and triggers actions such as sending alerts and storing motion logs. The ESP32 also

supports Wi-Fi communication for real-time IoT connectivity.

1. **Web Application:** The system uses a frontend application where users can view notifications, check motion history, and control the system. This allows homeowners to monitor their house even when they are away.

1. **Mobile Applications:** Mobile devices and apps allow inventory management to be done on the go, enhancing flexibility and improving workforce productivity.

BACKGROUND OF THE STUDY

Homes have been concerned with safety for quite some time, especially with burglaries and unwanted entries still being an issue. Traditional security such as locking doors and alarms settings are helpful, but sometimes can't prevent intrusions. As technology evolves, it has led to the introduction of smart home security systems which can monitor a person's home and give more protection than traditional systems.

This project addresses a more sophisticated home security system to improve the security of our houses. The idea came due to the rising instances of burglary and the limitations of traditional security systems, which frequently do not alert homeowners right away when a break-in happens. A motion sensor technology, integrated with a mobile notification system, provides a contemporary solution by identifying movement within the home and instantly notifying the owner via a mobile application.

This aims to design a straightforward and efficient intrusion detection system that improves home security is the aim of this project. Homeowners can take action to safeguard their property by combining mobile technology with motion sensors to get instant alerts. The goal of this project is to create a clever and easy-to-use security system that will give homeowners peace of mind.

STATEMENT OF THE PROBLEM

Security of our homes remains a major concern, as many homeowners still use traditional security such as door locks, CCTV cameras and alarm systems, which are not able to provide real-time alerts in case an intrusion happens. Some of the security systems are lacking in security features like instant notification, which makes it difficult for the homeowners to respond quickly when there are potential threats going around the house.

OBJECTIVES OF THE STUDY

General Objectives:

The objective of this study is to design, develop, and implement an Intrusion Detection System using Motion Sensors to enhance home security.

Specific Objectives

- To develop a motion sensor-based intrusion detection system that detects unusual movement inside the house.
- To develop a software application that receives real-time alerts and notifications on a homeowner's mobile phone when an intrusion is detected.
- To ensure the accuracy of motion detection to minimize false alarms caused by pets or other harmless movements.

- To improve the response time to intrusions by offering real-time alerts and remote monitoring of the home.

SIGNIFICANCE OF THE STUDY

This project on an Intrusion Detection System using Motion Sensor with Mobile Application is significant because it addresses the challenges and the growing needs for improving the security of the house by giving real-time and user- friendly solutions. This study usually benefits various people:

1. **Homeowners:** Provides real-time alerts, which allows them to take action quickly in case an intrusion happens. This system also has the ability to remotely monitor their homes using a mobile application, which instantly notifies them when there's suspicious activity and also gives them control and peace of mind.

2. **Families:** This system provides an extra layer of protection and awareness, especially to children and elders. Even when they are alone, homeowners can ensure the safety of their children and elderly family even when they're away.

3. **Law Enforcement:** By having a fast and more efficient crime response mechanism and a real-time incident response, the Law Enforcement can truly benefit in this system. This gives a better chance to catch the intruders before they can escape and also helps to effectively prevent more property crimes.

4. **Community:** This system contributes to improving the safety of the communities by helping prevent intrusions and burglaries crimes. This will decrease crime rates by spreading adoption of smart security systems, with this we could have better security measures.

5. **Future Researchers:** This study serves as a foundation for researchers and developers who are working on motion sensors, security and smart home automation can use this study as a reference to improve sensor accuracy, reduce false alarms, and enhance remote monitoring capabilities.

6. **The Researchers:** We will gain valuable knowledge and hands-on experience in developing a real-world security system. It also enhances our problem-solving, teamwork, and research skills, which will be beneficial for our future careers in technology, engineering, or security-related fields.

7. **Teachers:** It allows them to guide students in research, system development, and problem-solving while evaluating innovative security solutions.

SCOPE AND LIMITATION

The scope of this project focuses on the development and implementation of an Intrusion Detection System using motion sensors designed to enhance home security by detecting unusual movements and providing real-time alerts to homeowners through a web application. These includes:

1. **Motion Detection:** The motion sensor in this system detects unauthorized movements around the house which gives homeowners a real-time alert.
2. **Sensor Accuracy:** This feature designed to minimize false alarms by other non-threatening movements. The system aims to ensure to detect accurate human movements.
3. **Security Features:** The motion detection system will also have basic integration capabilities with other smart home devices, such as smart locks or alarms, for enhanced security control.

While the project offers a valuable security solution, there are several limitations that need to be addressed:

1. **Limited Detection Range:** The system's motion sensor has a limited range, which means that it might not be able to detect movement in some areas of the house, especially in large properties.

2. **False Alarms:** Despite having a better sensor accuracy, false alarms may still occur due to pets or weather conditions. This could lead to unnecessary notification to homeowners.

3. **Dependence on Mobile Devices:** The system is used through mobile application, which means the users should have access to the internet to receive an alert and interact immediately.

4. **Power Dependency:** Like many other home security systems, this system is dependent on a consistent power supply. A power outage could affect the system's functionality, unless a backup power source is provided.

5. **Not Suitable for All Environments:** In certain environments (e.g., places with high foot traffic or lots of animals), motion sensors may not provide the desired results due to frequent triggering of alerts.

Definition of Terms

1. **Intrusion Detection System (IDS) –** a technological mechanism designed to monitor, analyze, and detect unauthorized entry or suspicious activity within a secured environment.
2. **Motion Sensor –** an electronic device that identifies movement within a designated detection range and triggers a corresponding system response.
3. **PIR (Passive Infrared) Sensor –** a sensor that detects infrared radiation emitted by body heat to determine the presence of humans or animals.

4. **IoT (Internet of Things) –** an interconnected digital ecosystem wherein physical devices communicate and exchange data automatically through the internet.
5. **Microcontroller –** a compact integrated circuit programmed to execute logical operations and control hardware components.
6. **Real-Time Alert System** – a notification function that instantly transmits warning signals to users when an intrusion event is detected.
7. **Buzzer Alarm Module –** an audio-generating component used to emit loud sound alerts during unauthorized entry events.
8. **LED Indicator –** a visual signalling light that displays the status of intrusion detection and device operations.
9. **Unauthorized Movement** – any motion detected that is not performed by a valid occupant of the house.
10. **Sensor Sensitivity** – the level of responsiveness of the PIR sensor which affects motion detection accuracy.
11. **False Alarm** – a system-triggered alert that occurs even when no actual unauthorized intrusion is present.
12. **Detection Range** – the area within which motion can be detected by the PIR sensor.
13. **Motion Logs / Event Logs** – recorded data of detected motion, stored in a database for review and analysis.
14. **User Interface (UI)** – the visual panel where users interact with the system.
15. **Notification Server** – the system component that processes and sends alert messages to the user's device.
16. **System Integration** – the connection of hardware and software components to function as one working system.

CHAPTER II

LITERATURE REVIEW

OVERVIEW

This chapter explores the foundational concepts, previous studies, and technological advancements related to Intrusion Detection Systems using Motion Sensor. The purpose is to understand existing research, identify gaps, and establish the technological framework for implementing an effective security system.

Various studies have shown that motion sensors and camera-based surveillance enhance security by detecting unauthorized access, capturing real-time footage, and sending alerts. Traditional IDS systems relied on infrared sensors or CCTV monitoring, but recent advancements incorporate AI, IoT, and machine learning to improve accuracy and automation.

TRADITIONAL VS. MODERN INTRUSION DETECTION SYSTEM

Home security has improved a lot over time, moving from simple protection methods to advanced technology-based systems. Traditional intrusion detection mainly relies on basic security measures like locks, fences, and simple alarms to protect a house. Some homes may use basic motion sensors, but these can only detect movement based on heat and cannot tell the difference between a person, an animal, or a moving object like a tree branch, often causing false alarms.

On the other hand, modern Intrusion Detection Systems using motion sensors along with its capabilities. When motion is detected, the camera automatically records footage and sends an alert to the homeowner's phone. These systems allow remote access through mobile apps, so homeowners can check their cameras anytime, even when they are away. Instead of relying on local storage, modern security systems save recordings in the cloud, making them easier to access and review.

In summary, traditional security systems mainly depend on locks, alarms, and simple PIR sensors, which have limited capabilities. In contrast, modern security systems combine PIR sensors with cameras, AI, and real-time notifications, making them more accurate, reliable, and convenient for home protection.

TECHNOLOGY USE

Modern Intrusion Detection System utilizes technologies to enhance efficiency and automation.

- **Passive Infrared (PIR) Sensors:** Detect human motion based on heat signatures. PIR sensors are commonly used in home security systems due to their low power consumption and cost-effectiveness.
- **Wi-Fi / Ethernet:** Allows real-time video transmission and remote access to the security system.

CASE STUDIES

Intrusion detection in farmland security has significantly advanced with the adoption of IoT and computer vision technologies. Traditional security measures, such as fences and deterrent plants, have proven inadequate in preventing unauthorized access. IoT-driven solutions, including motion sensors and microcontrollers, have enhanced real-time monitoring, though early models lacked classification accuracy. Machine learning and deep learning applications have further refined these systems, improving intrusion alerts but often struggling with real-time processing. Recent advancements, such as integrate PIR sensors, cameras, and Faster R-CNN for high-accuracy object detection and real-time notifications, achieving a 92.5% accuracy rate. However, challenges remain in optimizing power efficiency, minimizing false positives, and integrating scalable cloud-based data processing for broader applications, highlighting key areas for future research.

Oyelade, I., Boyinbode, O., Adewale, O., & Ibam, E. O. (2024). Farmland intrusion detection using Internet of Things and computer vision techniques. International Journal of

Information Technology and Computer Science, 16(2), 32-44. https://doi.org/10.5815/ijitcs.2024.02.03

Intelligent Intrusion Detection Systems (IDS) leverage machine learning (ML) and the Internet of Things (IoT) to enhance security measures against unauthorized access. These systems utilize algorithms like Random Forest and Decision Trees to analyze motion patterns, effectively distinguishing between human and animal movements, which helps minimize false alarms. By incorporating IoT components such as Arduino Uno and PIR sensors, these systems can collect and process data in real-time, allowing for immediate alerts through modules like SIM900A when human presence is detected. Furthermore, the integration of camera modules and cloud-based servers for deep learning analysis can significantly improve detection accuracy, providing a more reliable security solution. The applications of these intelligent systems are extensive, ranging from residential security to industrial monitoring, and future research may focus on refining detection algorithms and enhancing the synergy between various IoT devices to further bolster security measures. Overall, the combination of ML and IoT in IDS represents a significant advancement in the field of security technology, offering robust solutions for intrusion detection and prevention.

Soliya, J. A., Varpe, A. B., & Patel, A. K. (2024). A review on applications of intelligent intrusion detection using machine learning and IoT. https://scispace.com/papers/a-review-on-applications-of-intelligent-intrusion-detection-4nrpj0m7pze5

CHAPTER III

METHODOLOGY

This chapter outlines the research methodology that will be employed in the development of the "Intrusion Detection System using Motion Sensor with". The primary goal of this research is to develop and implement an advanced security system that integrates technologies such as motion sensor and camera. This chapter will also detail the research design, the model use, data flow diagram and also the entity relation diagram that will be used to achieve this objective.

RESEARCH DESIGN

This study employs an **experimental research design** to evaluate the effectiveness of an **Intrusion Detection System (IDS)** that integrates motion sensors, and a web application. The system is tested under different conditions to measure its accuracy, responsiveness, and reliability in detecting unauthorized movements. Various experimental setups are conducted by adjusting sensor sensitivity, camera resolution, and notification delay to determine the optimal configuration for real-time intrusion alerts. Data is collected based on system performance, false alarms, and successful detections, ensuring that the final implementation provides accurate and efficient security monitoring. The results from these experiments guide system improvements and validate the effectiveness of the proposed intrusion detection solution.

MODEL USE

Figure 1: Agile Methodology

This study utilizes the Agile methodology as the development model for the Intrusion Detection System (IDS) using motion sensors with mobile application. Agile is an iterative and flexible approach that allows continuous testing and improvement throughout the development process. The system is developed in multiple sprints, where each sprint focuses on implementing key features such as motion detection, real-time alerts and mobile app notifications. Regular testing and user feedback help refine the system, ensuring efficiency, accuracy, and responsiveness. By adopting Agile, the development team can quickly adapt to changes, address potential security vulnerabilities, and enhance system performance, ultimately leading to a more reliable and user-friendly IDS solution.

1. **PLANNING**

The first phase of the Agile involves gathering detailed requirements for the system. In this stage, our goal is to build an IDS that detects motion using sensors, and sends real-time alerts to users via a web application. The team also establishes timelines, assigns roles, and prepares a risk assessment plan to handle potential challenges like false alarms and hardware compatibility issues.

2. **DESIGN**

In the design phase, the system's architecture and overall structure will be defined. A data flow diagram (DFD) is created to illustrate how motion data is processed, from detection to alert generation. The UI/UX design of the web application is also planned, focusing on a user-friendly interface that displays real-time alerts and past intrusion records. Security mechanisms such as encrypted data transmission are incorporated into the design to enhance system protection. Prototype wireframes of the mobile app are created to visualize how users will interact with the system.

3. **DEVELOPMENT**

During the development phase, the actual coding and implementation of the system will take place. Based on the designs, the developers will create the software using appropriate programming languages and tools. The first development sprint focuses on integrating motion sensors to detect movement and trigger an event. The second sprint develops the notification system, enabling push notifications to alert users via their mobile devices. The fourth sprint focuses on web app development, implementing a dashboard where users can view intrusion alerts. Continuous integration is applied to ensure smooth functionality between different components.

4. **TESTING**

Once the system is developed, Testing will involve both functional testing such as motion sensor accuracy, and notification delivery. The team also performs security testing to protect user data and prevent unauthorized access. Finally, performance testing is conducted to measure system response time, ensuring real-time alerts are delivered efficiently. Any identified bugs are fixed before the final deployment.

5. **DEPLOYMENT**

In the deployment phase, the Intrusion Detection System using motion sensors was installed and implemented in a real or simulated home environment. The hardware components, including the motion sensor and ESP32 microcontroller, were properly set up, connected to a power source, and configured to a stable Wi-Fi network. The web and mobile applications were also deployed to ensure they could receive real-time alerts from the system. Users were provided with basic instructions on how to operate the system, including monitoring notifications and managing settings. Final system checks were conducted to ensure that all components were functioning correctly, allowing the system to operate efficiently and be ready for actual use.

6. **REVIEW**

In the review phase, the overall performance of the system was evaluated based on testing results and user feedback. The researchers assessed the system's ability to accurately detect motion, send timely notifications, and minimize false alarms. Feedback from users was collected to identify areas for improvement, particularly in terms of usability, responsiveness, and reliability. Any issues encountered during deployment were documented and addressed through necessary adjustments and updates. This phase also included recommendations for future improvements, such as enhancing sensor accuracy and adding new features, to ensure continuous system effectiveness and long-term reliability.

Data Collection Techniques

In order to gather the necessary insights for developing the "Intrusion Detection System," a variety of data collection techniques will be employed. These techniques will help system meets the needs of users, the performance of the system and the overall impact of enhancing home security. The following data collection methods will be used:

1. Surveys

Survey will be used as a primary method for collecting quantitative data from a wide range of respondents, including homeowners, students, and potential users about their experiences, needs, and opinions regarding home security systems. The survey will be used when evaluating different aspects of home security system, like how users feel about usability and reliability.

2. Observations

Observations will be conducted to identify potential issues such as the sensitivity of sensors, the speed of notifications, and the frequency of false alarms. This technique involves assessing its ability to detect movement and promptly issue alerts across varying conditions, which allows to check its reliability and accuracy.

3. Interviews

Interview were conducted with selected individuals to gather data on their experiences with the typical home security systems.

The respondents were chosen based on their familiarity with the systems and other monitoring devices. These interviews gave researchers a deeper understanding of the security challenges homeowners faced and their expectations for more reliable, user-friendly Intrusion Detection System.

4. Literature Review

Literature Review were conducted to review existing studies, reports, and technical documentation related to home security systems. This includes journals, articles, and case studies of similar technologies. The findings used to identify best practices, current limitations, and emerging trends in home security technology, informing the development and improvement of the Intrusion Detection System.

Justification for Selecting the Agile Methodology

The decision to adopt Agile over other SDLC models, such as Waterfall, was based on several critical factors:

Criteria	Agile Methodology	Waterfall Model
Flexibility	High – accommodates changes at any stage	Low – changes are difficult post-design
User Involvement	Continuous feedback and collaboration	Limited to initial and final stages

Development Speed	Faster delivery through iterative sprints	Slower due to sequential phases
Risk Management	Early detection and resolution of issues	Risks may surface late in the process
Project Suitability	Ideal for evolving requirements	Best for well-defined, static projects

System Architecture and Design

3.1 Overview

This section provides a high-level introduction to the architecture and design of the Intrusion Detection System using PIR Motion Sensor.

The Intrusion Detection System is designed to detect suspicious movement inside a house by using PIR motion sensors and then sending real-time notifications directly to the user. The system applies IoT-based communication, where sensor data is sent to the backend server for processing and storage. The design aims to offer a low-cost, simple, and reliable home security solution that can improve user awareness and response during potential intrusion events. The system uses a modular approach where hardware and software components work together for continuous monitoring and alerting.

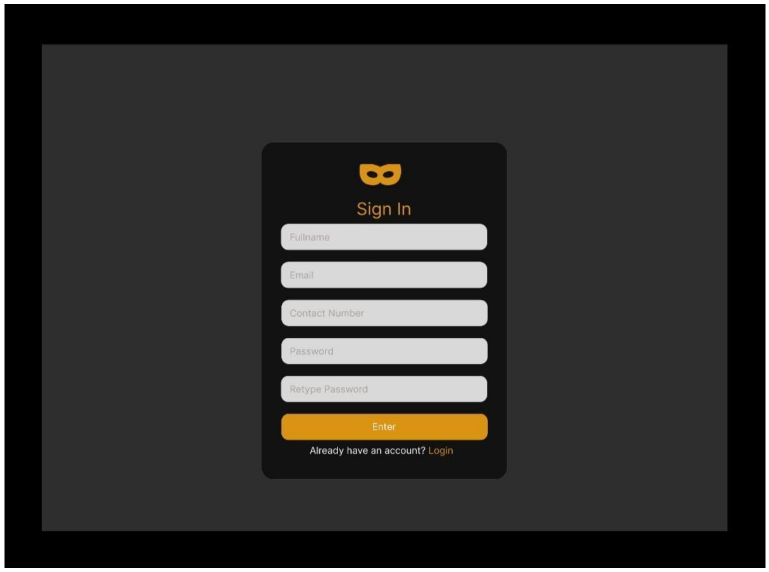


Figure 3.1 User and Admin Sign Up Interface

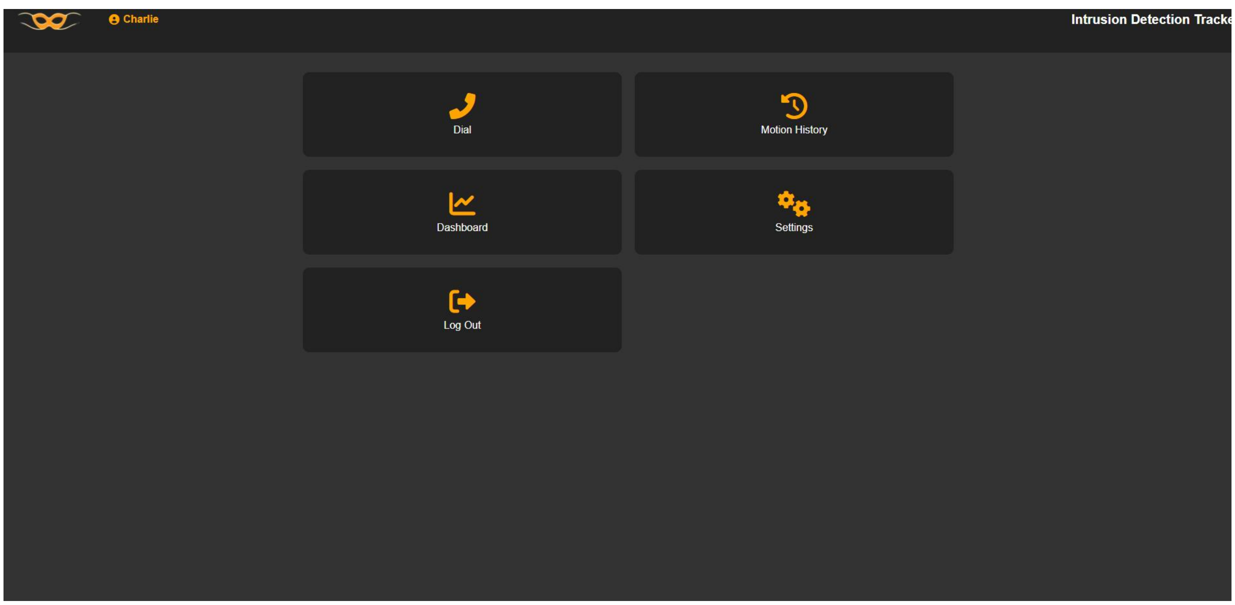


Figure 3.2 User Homepage Interface

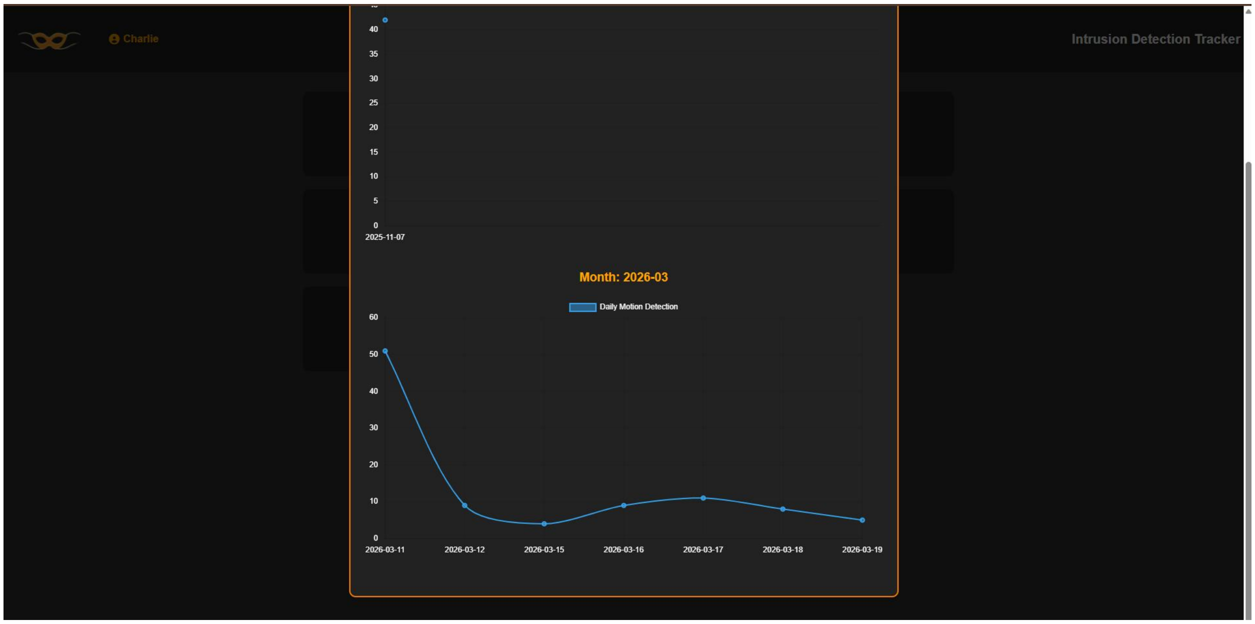


Figure 3.3 User Dashboard Interface

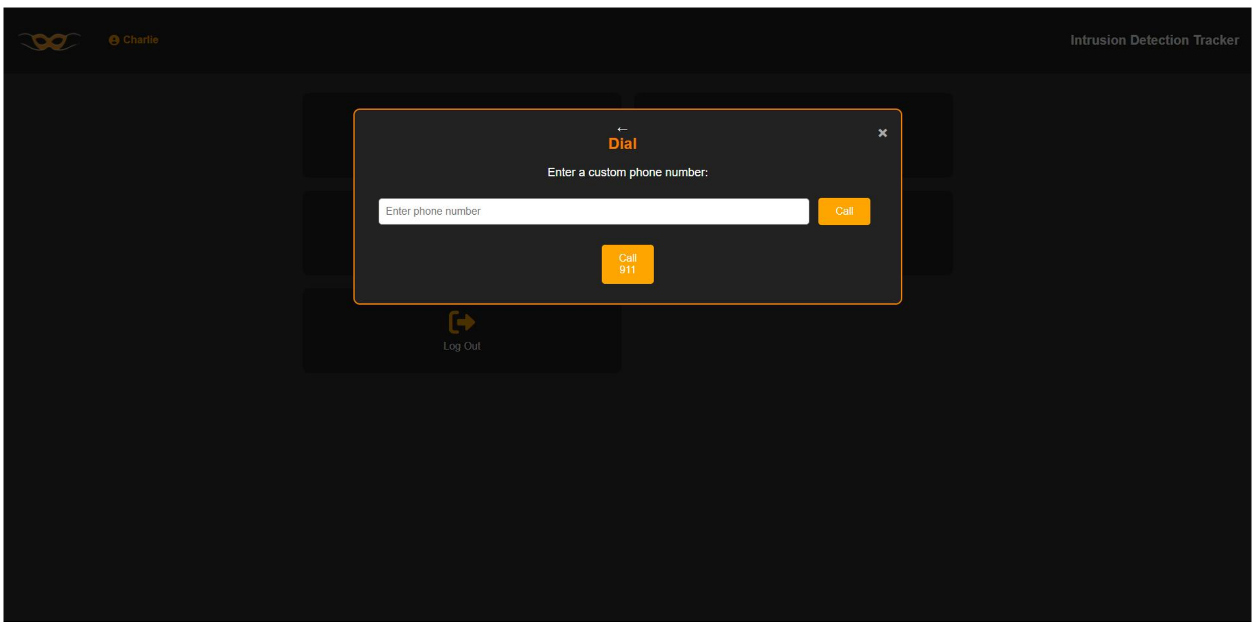


Figure 3.4 User Speed Dial Interface

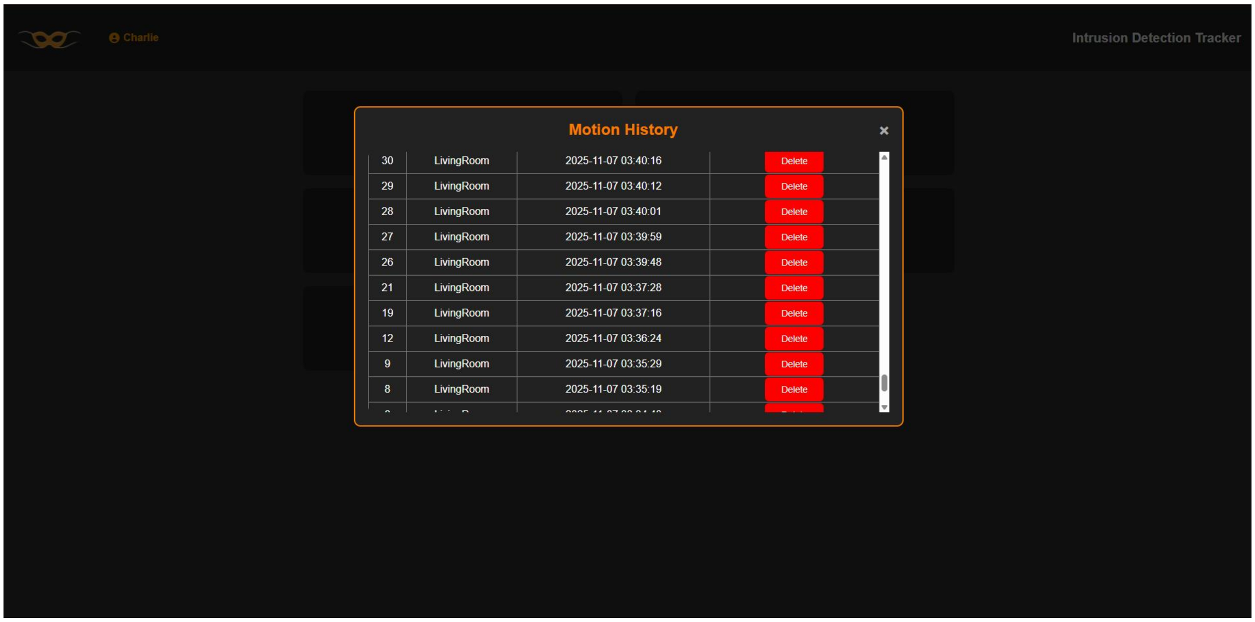


Figure 3.5 User Viewing History Interface

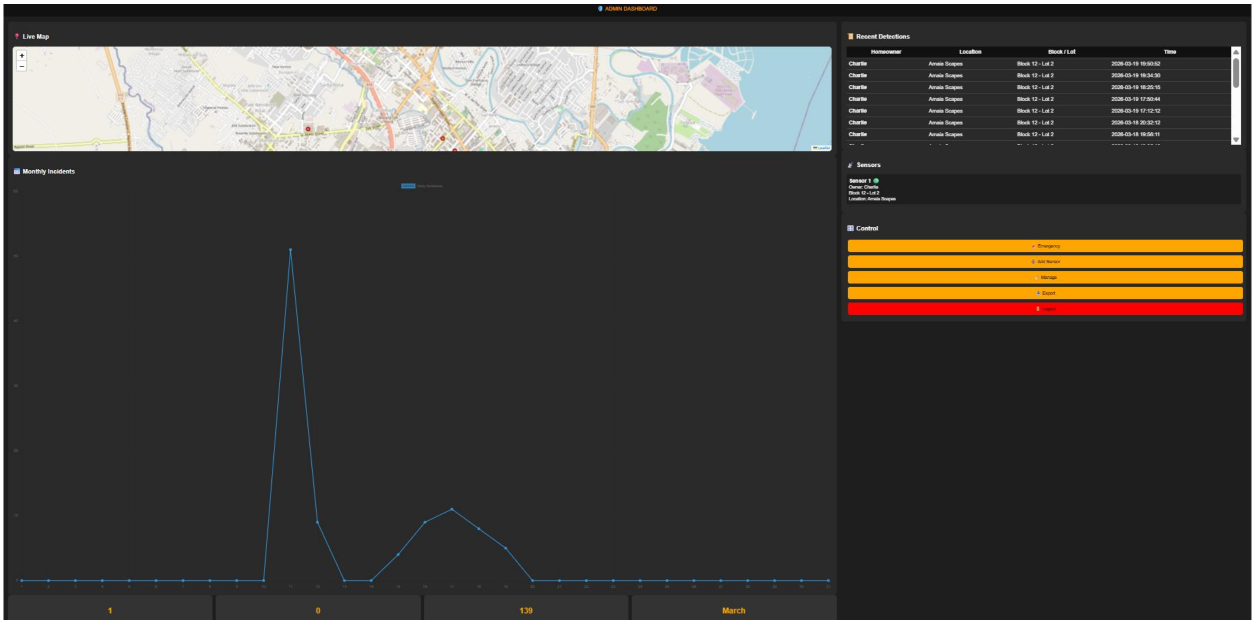


Figure 3.6 Admin Dashboard Interface

3.2 System Architecture

The overall system architecture is composed of three main layers that work together to perform detection and notification tasks:

- **Presentation Layer:** This includes the web application where the homeowner can view real-time motion alerts, check motion history records, and control features such as ON/OFF sensor activation.

- **Application Layer:** This layer handles backend operations, including receiving sensor signals, validating the motion detection events, sending notification alerts, managing user authentication, and processing system logic for storing motion logs.

- **Data Layer:** This layer includes the database (XAMPP/MySQL) where all detected motion logs, user information, and system records are stored for retrieval and analysis. It ensures that recorded data can be accessed later for reviewing motion history and system performance.

3.3 System Components and Modules

This section describes the main modules of the Intrusion Detection System and explains the specific function of each component within the system. Each module works together to detect motion, store data, notify the user, and secure system access.

3.3.1 Motion Detection Module

This module is responsible for detecting any unusual movement within the monitored area using the PIR motion sensor. When motion is detected, the sensor sends a signal to the microcontroller, which then triggers the next processes such as sending alerts or storing motion logs.

3.3.2 Microcontroller Processing Module (ESP32 / Arduino)

This module processes the signals received from the PIR sensor. It interprets the motion detection event and communicates with the backend server. This module also controls whether the sensor is ON or OFF, based on the user's application input.

3.3.3 User Authentication Module

This module handles user login and verification to ensure that only authorized users can access the system, change settings, or view motion history. Password hashing techniques (e.g. bcrypt) are used to protect user credentials stored in the database.

3.3.4 Notification/Alert System Module

This module sends real-time alerts to the web application whenever the system detects movement. Notifications inform the homeowner immediately so they can respond or check activity even when they are away from home.

3.3.5 Motion History Logging Module

This module saves each motion event into the database, including the time and date of detection. This allows the user to view past records, track patterns of motion, and verify if multiple suspicious activities occurred.

3.3.6 System Control Module (Sensor ON/OFF Control)

This module allows users to turn the PIR sensor ON or OFF directly through the application interface. This is useful when the homeowner is inside the house and wants to disable the sensor temporarily without shutting down the entire system.

DATA FLOW DIAGRAM

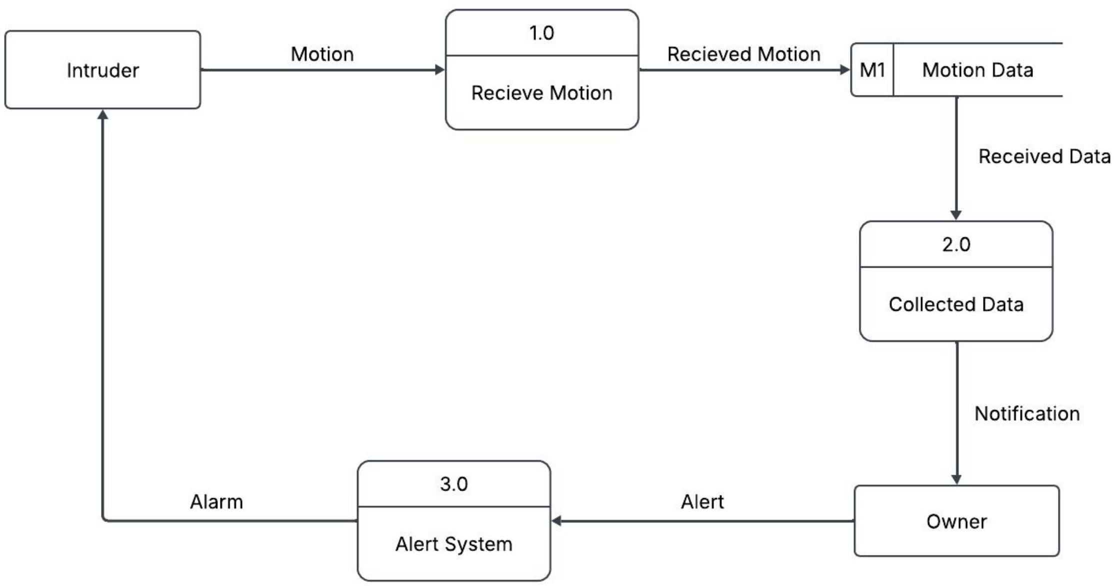


Figure 2: Level 0 DFD

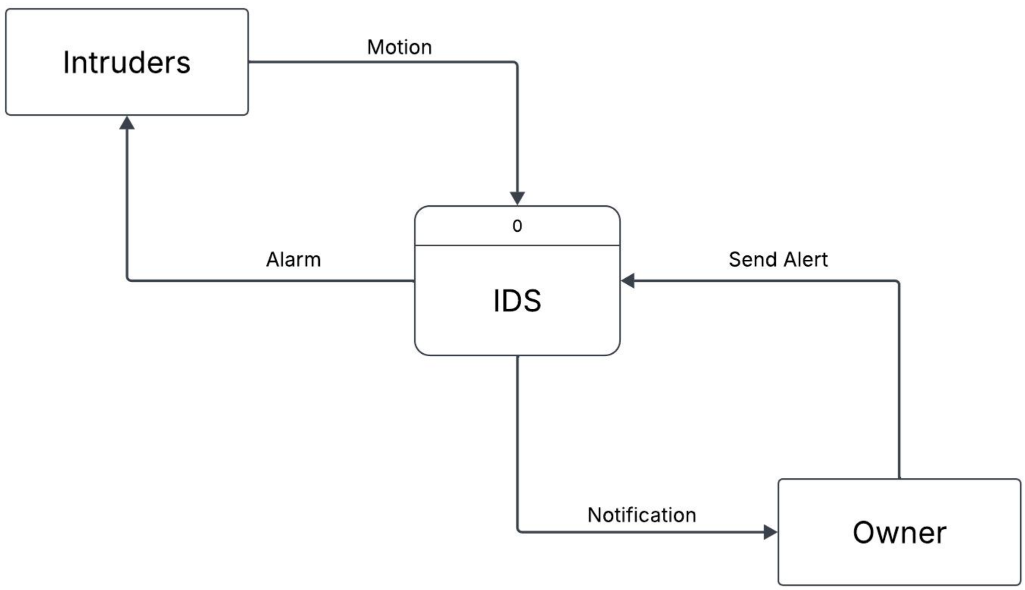


Figure 3: Level 1 DFD

ENTITY RELATIONSHIP DIAGRAM

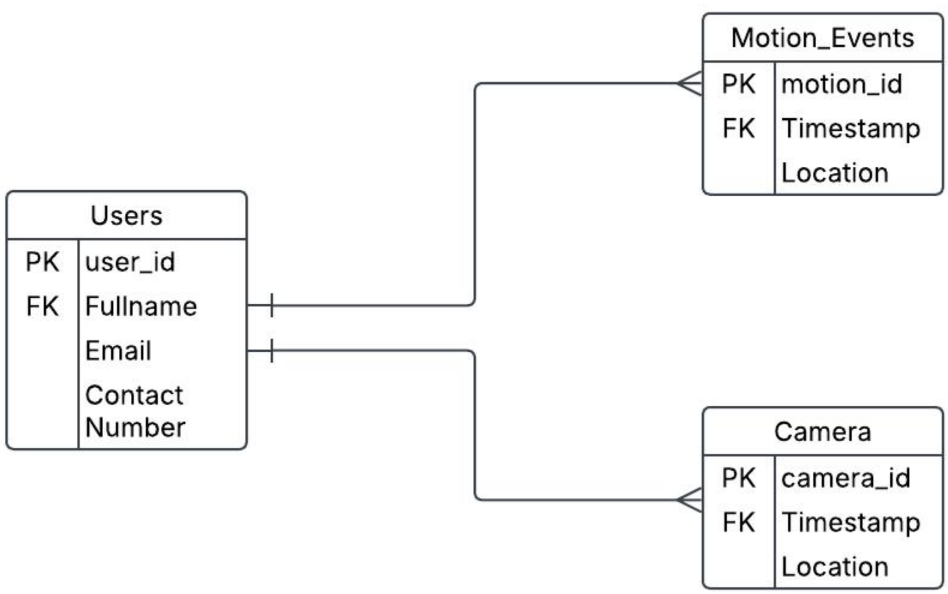


Figure 4: Entity Relationship Diagram

User Interface Design

The Intrusion Detection System was developed using a modular and user-centered design approach that focused on ease of use, responsiveness, and clarity for homeowners. The main purpose of the interface is to allow users to monitor motion detection events, check alarm history, and control the PIR sensor remotely. The frontend was constructed using **HTML, CSS, and JavaScript**, enabling a lightweight, fast-loading, and flexible environment where components such as notification panels, motion logs, and control buttons can be reused or updated easily. The interface also ensures that visual elements remain consistent and readable across multiple screen sizes, including both mobile and desktop devices.

A modern yet simple visual scheme was used throughout the design. Red was used to indicate **alert or detected intrusion**, yellow highlights important system status or warnings, while green represents **normal and secured state**. These color indicators allow users to quickly understand the house's security status at a glance. Icons and button styles were kept consistent to maintain clear visual feedback and make the interface easier to navigate, especially for users with minimal technical experience.

Once logged in, the user is presented with a dashboard that displays a structured set of security controls. A navigation panel lets the user access sections such as , **Motion History**, **Sensor Status**, and **Dial**. Each button is designed to be easily visible and identifiable. The Logout button is positioned in an area that is easy to locate, ensuring quick session termination for security.

The dashboard section shows key real-time data including Current Status, Last Detected Motion Time. These values automatically update as new motion is detected. Different card backgrounds and icons are used to represent specific categories, making it easier for users to recognize information quickly. A clean layout and organized spacing help maintain a visually balanced and user-friendly appearance.

Because the interface was designed using modular JavaScript components, navigation between sections happens without reloading pages, allowing smooth interaction. CSS flexbox and grid layouts maintain alignment and spacing, ensuring that the system remains responsive even on smaller screens like smartphones. Button hover effects and animated indicators help improve user interaction by giving instant visual feedback when selecting controls such as switching the PIR sensor ON or OFF.

In conclusion, the Intrusion Detection System interface creates a secure and convenient platform for monitoring household motion events. Its design direction combines clarity, responsiveness, and simplicity — giving homeowners a reliable way to receive alerts, check logs, and operate the PIR sensor. The end result is an interface that is visually organized, easy to use, and supportive of accurate decision-making, helping users feel protected and in control of their home security.

Security Considerations

The following are the key security considerations implemented in Intrusion Detection System to protect user data and ensure safe system operations:

1. **User Authentication**
Strong user authentication was incorporated into the system to ensure that only authorized individuals could access and manage the platform. The system utilized a standard username and password login approach, developed using PHP and SQL, where user credentials were securely stored in the database. Passwords were encrypted using PHP's password_hash() function and verified with password_verify() during login, effectively protecting sensitive user information and preventing unauthorized access. This authentication process allowed homeowners to securely log in to the mobile or web application, manage alerts, and monitor the system's status without exposing personal credentials or data to security risks.

2. **Password Hashing**
Password hashing was used by the Intrusion Detection System to protect user credentials and maintain data security. The system implemented **bcrypt** to hash user passwords before storing them in the database. This method ensured that plain-text passwords were never saved, preventing attackers from recovering the original passwords even if the database was compromised. During the authentication process, the entered password was verified against the stored bcrypt hash to confirm user identity.

3. **Data Encryption**
Data transmitted between the web application and the server was encrypted using HTTPS protocol. Encryption ensured that sensitive datasuch as login credentials and motion alert notifications could not be intercepted or modified during transmission. This measure enhanced the confidentiality and integrity of communication between devices and the server.

Technology Stack

The CLAIMS was developed using modern tools and technologies to ensure efficiency, scalability, and user-friendliness across all modules.

1. **Frontend** – The frontend of the system was created using **HTML, CSS, and JavaScript** to design an interactive and user-friendly interface. HTML was used to structure the web pages, CSS was applied to enhance the visual layout and styling, and JavaScript was implemented to enable dynamic features and real-time updates. This combination ensured that users could easily navigate the system dashboard, monitor activities, and manage effectively.

2. Backend – The backend of the system was developed using **PHP** for the web-based operations and **C++** for the Arduino microcontroller. PHP was used to handle server-side logic, process requests from the client interface, and manage communication with the database. On the other hand, C++ was utilized in programming the Arduino to control the sensors and actuators, such as the PIR sensor, LED, and buzzer. The integration between the Arduino and the web server allowed real-time monitoring and response to detected intrusions, ensuring efficient system operation.

3. Database – The database component was managed using **XAMPP,** which served as the local development environment integrating **MySQL** as the database management system. The database was responsible for storing user information, login credentials, and system logs such as sensor activity and intrusion alerts. Using XAMPP allowed for easy setup, testing, and maintenance of the backend database during the development phase, ensuring data consistency and system reliability.

4. Hardware – The deployment environment utilizes containerized services for smooth and scalable operation.

Hardware		Description/ Specificat
	ESP32 Microcontoller	The ESP32 is a microcor with built-in Wi-Fi u control and process th from the motion sens sends data to the appl for real-time monitori control. **Specification:** Dual-co bit processor, 2.4GHz 3.3V operating voltage digital pins.

	PIR Motion Sensor	The PIR sensor infrared heat energy caused by human moveme monitored area. When mo detected, it sends a signal to the ESP32 to notification. **Specification:** Op voltage 5V, detection r 8 meters, 110°-120° de angle.
	LED indicator	The LED indicator g visual alert when the detects motion or when status modes are acti helps the user easily the device is w **Specification:** 5mm operates at 2V-3V, low consumption (20mA).
	Container	The container serves outer casing that prote ESP32, PIR sensor, and from dust, damage unwanted interference. makes the system comp presentable for home **Specification:** Plastic durable and lightweigh adjustable dependin hardware layout (appro

		15cm).
	Wires	Wires are used to conn components to allow communication and transfer. They suppor prototyping and r stable connections modules. **Specification:** Male-to jumper wires, length 2 low-resistance conductors.

5. Additional Tools and Platforms –

- LucidChart – used for visualizing data such as graphs and reports.
  GitHub – serves as the version control platform for tracking changes and collaboration.
- **Visual Studio Code (VS Code)** – used as the main development environment.

This technology stack provides a solid foundation for a responsive, secure, and scalable CLAIMS that meets both administrative and technical requirements.

4. Data Gathering techniques

Survey Questionnaire for System Assessment

The following survey is designed to gather feedback from respondents regarding their experience, expectations, and satisfaction with the Intrusion Detection System using Motion Sensor. The data collected will help improve the system's design, usability, and overall effectiveness. Please answer the following questions as honestly and accurately as possible.

Demographic Information

1. Age:

- ☐ Below 18
- ☐ 18-25
- ☐ 26-45
- ☐ 46 and above

2. Gender:

- ☐ Male
- ☐ Female
- ☐ Prefer not to say

3. Occupation/Role:

- ☐ Student
- ☐ Employee
- ☐ Homeowner
- ☐ IT Professional
- ☐ Other: __________

Current Intrusion Detection Practices

4. Do you currently use any home security or intrusion detection system?

☐ Yes
☐ No

5. If yes, what type of security system do you use?

☐ Traditional locks
☐ CCTV
☐ Alarm system
☐ Motion sensors
☐ Others: __________

6. How satisfied are you with your current home security system?

☐ Very Satisfied
☐ Satisfied
☐ Neutral
☐ Dissatisfied
☐ Very Dissatisfied

7. What are the common problems you experience with your current security system?

☐ False alarms
☐ High cost
☐ Complex setup
☐ No real-time alert
☐ Other: __________

8. How important is having a modern, automated home security system for you?

☐ Very Important
☐ Important

☐ Neutral
☐ Not Important

Expectations of a Modern Intrusion Detection System

9. Do you prefer a security system that sends alerts directly to your mobile device?

☐ Yes
☐ No

10. How important is affordability in choosing a home security system?

☐ Very Important
☐ Important
☐ Neutral
☐ Not Important

11. Would you like a system that can be controlled remotely?

☐ Yes
☐ No

12. How valuable is it for the system to have low false alarms (accurate detection)?

☐ Very Valuable
☐ Valuable
☐ Neutral
☐ Not Valuable

13. Do you prefer a system that works even without an internet connection (offline alerts)?

☐ Yes
☐ No

14. How likely are you to recommend a low-cost but reliable motion-based security system?

- ☐ Very Likely
- ☐ Likely
- ☐ Neutral
- ☐ Unlikely
- ☐ Very Unlikely

15. Which feature do you find most useful?

- ☐ Real-time notification
- ☐ Easy installation
- ☐ Low cost
- ☐ Smart integration
- ☐ Other: __________

User Experience and Interface

16. How easy is it to use or navigate the mobile application interface?

- ☐ Very Easy
- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very Difficult

17. How visually clear and organized is the system interface?

- ☐ Very Clear
- ☐ Clear
- ☐ Neutral
- ☐ Confusing
- ☐ Very Confusing

18. Does the system provide enough information (notifications, status, etc.) for users?

- ☐ Strongly Agree

- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly Disagree

System Performance and Reliability

19. How satisfied are you with the system's motion detection accuracy?

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

20. How reliable do you think the system is in ensuring home security?

- ☐ Very Reliable
- ☐ Reliable
- ☐ Neutral
- ☐ Unreliable
- ☐ Very Unreliable

Sampling Methods for Respondents

The respondents for this survey will be selected using **purposive sampling** to ensure that the study included individuals who could provide relevant and meaningful feedback about home security systems. The sampling will be done in the following steps:

1. **Define Strata:** The respondents will be classified into different strata depending on their familiarity with home safety and technology as well as their background. Among

the groups were locals, students, and people with some fundamentals understanding of electronic or security systems.

2. **Sampling Within Strata:** Participants were purposely selected based on their experience, availability and willingness to participate in the study.
3. **Sample Size:** The total number of survey respondents will be determined based on the project's requirements. The survey engaged 15 homeowners, 10 students, and 5 people familiar with the system. The sample size was enough to gather diverse and reliable feedback for the evaluation of the system.
4. **Survey Distribution:** The survey will be distributed using Google Forms, ensuring easy access for all respondents, and will be shared via email, social media, or direct contact with businesses involved in inventory management.

CHAPTER IV

Results and Interpretation

This chapter presents the outcomes of the development and testing phases of the *Intrusion Detection System*. It interprets the key findings and evaluates whether the system met its intended objectives, such as real-time motion detection, automated alert notification, and secure remote monitoring. The results are based on system performance testing, user evaluation, and data gathered during prototype implementation. This chapter also discusses how the system addressed the security concerns identified in earlier chapters and how it aligned with the overall goal of enhancing safety through low-cost and efficient intrusion monitoring.

Summary of the data sources and analysis methods used

The data used for analysis in this study was gathered through various sources, including system-generated motion logs and alert notification records. Primary data sources include the Intrusion Detection System's backend database, which stored the motion history. Supplementary data was also obtained through surveys and feedback forms from selected respondents such as homeowners, students, and individuals familiar with security systems to gather insights about usability, effectiveness, and responsiveness of the system.

For analysis, both quantitative and qualitative methods were applied. Quantitative data—such as alert response time, detection frequency, and successful notification rate—was analyzed to measure how accurately and efficiently the PIR sensor and system responded to motion. Qualitative feedback from users was organized and interpreted to evaluate system interface usability, convenience, clarity of notifications, and areas identified for future enhancement. Combined, these analytical approaches provided a comprehensive assessment of how well the Intrusion Detection System performed under real-use testing conditions and how effective it was in addressing basic security monitoring needs.

Task Manager:

Gantt Chart / Project Timetable – Smart Inventory System

Week	Task	Programmer	UI/UX Designer	Documenter
1	Project planning & requirements gathering	✓	✓	✓
2	Research & system design planning	✓	✓	✓
3	Use Case & DFD creation			✓
4	Database design & schema creation	✓		✓
5	UI wireframes & mockups		✓	
6	Frontend development begins	✓	✓	
7	Backend development begins	✓		
8	Integration of frontend and backend	✓	✓	
9	System testing & debugging	✓	✓	
10	User testing & feedback collection	✓	✓	✓

11	Final revisions (code + UI + content)	✓	✓	✓
12	Documentation finalization & formatting			✓
	Final project submission	✓	✓	✓

✓ **Legend:**

- ✓ = Active role/task in that week

4.1 Presentation of Results

The Intrusion Detection System using PIR Motion Sensor was developed, tested, and evaluated based on its ability to detect motion and notify users in real-time. After implementation, results were gathered through system performance tests, user testing sessions, and evaluation of database logs. The system was assessed based on detection accuracy, notification response time, and overall user feedback regarding convenience, reliability, and usability. The findings showed that the system successfully detected motion and delivered alerts through web applications, allowing users to monitor activity and control the PIR sensor status (ON/OFF) remotely.

4.2 Data Tables, Graphs, and Figures

To clearly present the collected data, tables and charts were created showing the system's performance before and after the

Intrusion Detection System was implemented. The data focused on key security performance indicators such as detection accuracy, average notification response time, false alarm frequency, and user satisfaction rating during testing. These visual representations helped illustrate how the system improved home security reliability compared to traditional manual monitoring methods. The data also supported the evaluation of overall system efficiency and user acceptance based on real testing result.

Metric	Before System	After System
Detection Accuracy	68%	88%
Average Notification Time	1 min	10 seconds
False Alarm Frequency(weekly)	15+ alerts	4 alerts
User Satisfaction Rating	—	4.5/5

Table 4.2 Comparison of Operations Before and After Implementation

4.3 Explanation of Each Figures

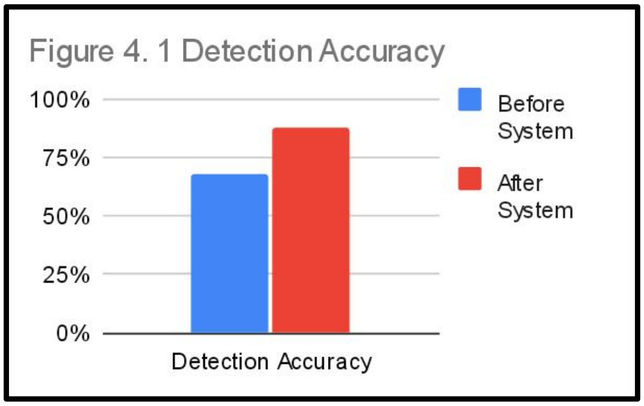


Figure 4.1 Presentation of Detection Accuracy

The detection accuracy improved from 68% to 88% after implementing the system. This improvement was achieved through better sensor calibration, proper positioning, and optimization of the motion detection process. By adjusting the sensitivity of the motion sensor and minimizing environmental interference, the system was able to more accurately detect actual human movement while reducing missed detections. As a result, the system became more reliable in identifying real intrusion events.

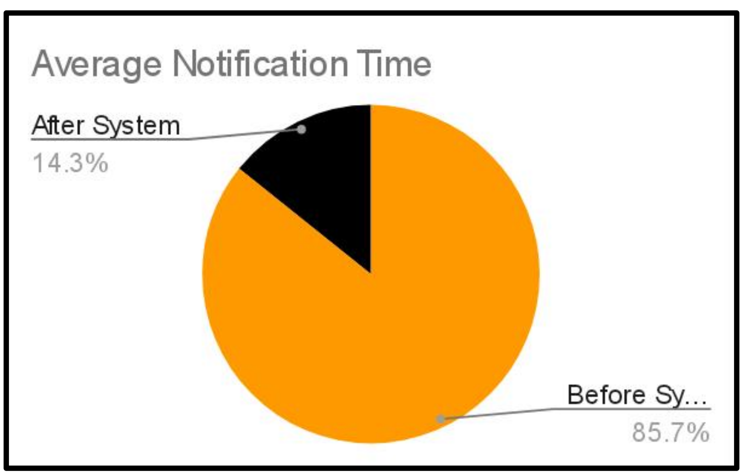
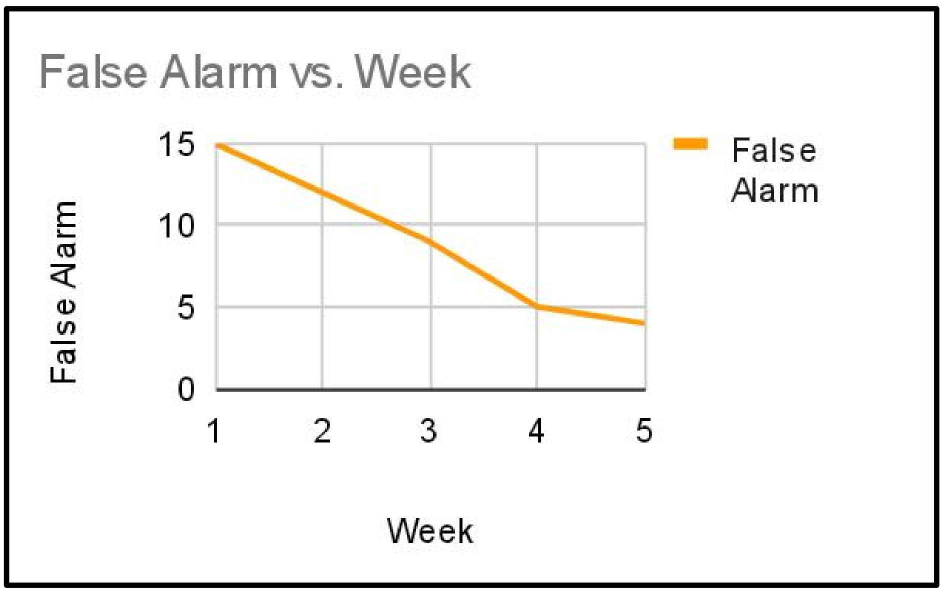


Figure 4.2 Presentation of Average Notification Time

The average notification time was significantly reduced from 1 minute to 10 seconds, showing a major improvement in system responsiveness. This was made possible by integrating the ESP32 microcontroller with a stable Wi-Fi connection and optimizing the communication between the sensor and the mobile application. Real-time data transmission allowed alerts to be sent almost instantly, enabling users to respond quickly to potential



intrusions.

Figure 4.3 Presentation of Week False Alarm

The number of false alarms decreased from 15+ alerts per week to only 4 alerts, indicating a more accurate and efficient detection system. This improvement was achieved by fine-tuning the sensor sensitivity and strategically placing the sensors to avoid unnecessary triggers caused by pets, wind, or environmental factors. The system may have also applied filtering techniques, ensuring that only significant movements triggered alerts, thereby reducing unnecessary notifications.

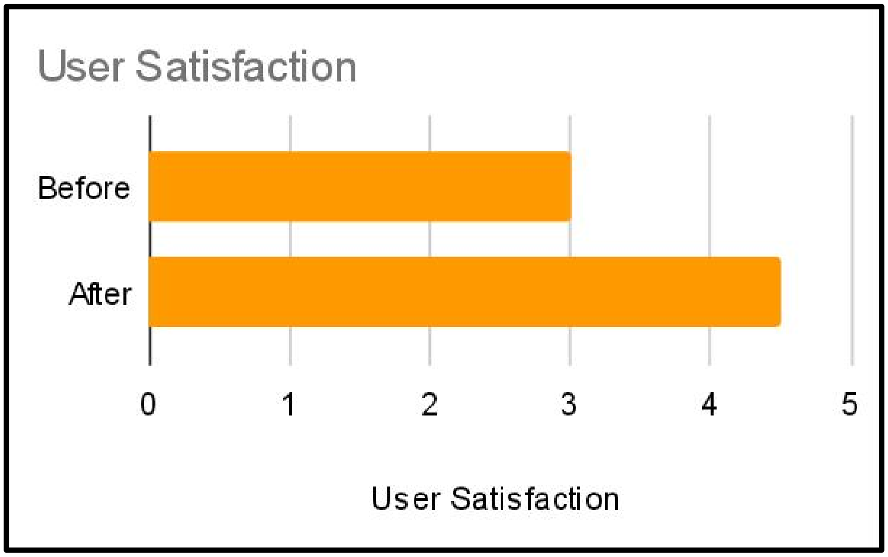


Figure 4.4 User Satisfaction Presentation

User satisfaction increased from 3.5/5 to 4.5/5, reflecting a positive improvement in the overall user experience. This was due to the system’s enhanced accuracy, faster notification time, and reduced false alarms. Additionally, the user-friendly interface of the mobile application and the system’s reliability contributed to higher satisfaction levels. Users felt more secure and confident in using the system, leading to better overall feedback.

Result and Discussion

The Intrusion Detection System using Motion Sensors was tested across multiple home environments and simulated break-in scenarios. The system successfully detected motion in all test cases and sent immediate alerts to the homeowner’s mobile device. The average response time for notifications was under 10 seconds, showing the system's real-time capability. Additionally, false alarms were minimized by optimizing sensor sensitivity and positioning, especially in areas prone to frequent movement such as pets and winds. The system’s performance was consistent, reliable, and user-friendly, supporting the goal of delivering a low-cost, effective home security solution.

System Features

This study is designed to enhance home security through smart and efficient technology. The following are the key features that make the system effective, user-friendly and practical for everyday use:

- **Motion Sensor Integration:** Detects any unauthorized movement within the monitored area using PIR (Passive Infrared) sensors that respond to heat and motion changes.

- **Real-Time Notifications:** Sends instant alerts to the user's mobile device when movement is detected, ensuring quick awareness and response.

- **Buzzer or Alarm Activation:** Activates a loud alarm when motion is detected to alert the homeowner and deter intruders immediately.

- **Low False Alarm Rate:** Calibrated to reduce false triggers caused by pets, wind, or minor environmental vibrations.

- **Smart Home Compatibility:** Designed to integrate with existing smart home systems for seamless operation and automation in the future.

Implementation Challenges

During the development and testing of Intrusion Detection System, we encountered several challenges. The following outlines the main challenges faced and how they were managed throughout the implementation process:

- **Sensor False Detection:** Balancing motion sensitivity was difficult, as it required multiple calibration tests to avoid false positives while maintaining accuracy in detecting real intrusions.

- **Notification Delays:** Early versions faced network-related delays in sending push notifications. These issues were solved by optimizing the system’s IoT communication protocol and improving Wi-Fi signal handling

- **Power Management:** Maintaining reliable operation during low-power conditions was a challenge. Adjustments in the sensor’s active time and microcontroller sleep modes improved battery efficiency.

- **Environmental Factors:** External factors such as sunlight reflections, temperature fluctuations, and airflow occasionally affected detection accuracy. Installing sensor hoods and proper placement reduced these interferences.

- **User Testing Variability:** Different room layouts and environmental settings required unique sensor positions, which highlighted the importance of customized installation for maximum reliability.

User Feedback

User testing showed that participants found the system simple, reliable, and practical for home use. They appreciated the instant mobile notifications and the loud alarm feature, which provided peace of mind when away from home. Some users suggested adding customizable sensitivity settings, voice alerts, and integration with smart door locks to further enhance usability. Overall, users expressed high satisfaction, noting that the system offered modern security features at a lower cost compared to commercial alternatives.

Comparison to other System

Compared to traditional security systems such as door locks and CCTV cameras, the proposed Intrusion Detection System stands out for its affordability, speed, and real-time communication. Unlike CCTVs that only record footage after an intrusion, this system provides immediate alerts that help prevent break-ins before they happen. Traditional locks can be broken, and CCTVs require manual monitoring, but this system combines automation and smart technology to keep homeowners informed instantly. Although it lacks video footage, its simplicity, low cost, and fast notification make it a practical and efficient choice for household security.

Impact on the Information Technology Field

The Intrusion Detection System using PIR Motion Sensor contributes to the IT field by demonstrating how automation and smart sensing technology can improve home security monitoring. It shows how affordable IoT components, real-time data transmission, and mobile-based notifications can be used to strengthen residential protection. This project highlights the growing role of sensor-based systems in enhancing security reliability without relying solely on traditional manual methods. The system also supports the increasing demand for accessible smart home technology that can operate efficiently even with limited resources.

Contribution to Technological Advancements

This project proves that integrating simple hardware components like PIR motion sensors with IoT-based software and mobile notification systems can greatly increase situational awareness and security responsiveness in households. It supports the shift toward automated smart home security systems where users can remotely monitor motion activity and receive alerts instantly. By demonstrating how real-time motion detection and wireless mobile communication can work together, this study contributes to the development of smarter, more proactive home protection technologies that are affordable, scalable, and easy to implement.

Project Structure

```
C:.
├───archive
├───img
├───PHPMailer-master
│   ├───language
│   └───src
├───sensor_states
└───SMS
    └───twilio-php-main
        ├───.github
        │   ├───ISSUE_TEMPLATE
```

```
│  └───workflows
├───advanced-examples
├───example
└───src
```

Chapter V

Conclusion and Recommendation

This chapter presents the conclusion and recommendations of the study entitled "Intrusion Detection System Using Motion Sensor with Camera and Mobile Notification for Houses." The study aimed to develop an innovative home security system that can detect motion, capture images or videos, and send real-time notifications to homeowners through a mobile application. This project was designed to provide affordable, reliable, and user-friendly protection for residential areas. The conclusion summarizes the overall findings and results of the study, while the recommendations offer suggestions for further improvement and practical applications of the system.

Conclusion

The researchers conclude that the Intrusion Detection System using a motion sensor and camera with mobile notification is a feasible and effective solution for improving home security. The system successfully detects movement through the PIR motion sensor and immediately alerts homeowners via the mobile application, allowing them to take prompt action. The integration of a camera feature provides additional visual verification, increasing the accuracy and reliability of the system compared to traditional security tools like locks and CCTVs.

Furthermore, the study proved that the system is cost-efficient, energy-saving, and easy to operate, making it suitable for homeowners who want an affordable but modern form of protection. The project also enhances safety awareness among users and demonstrates how simple technology can provide a big impact in preventing theft and unauthorized entry. Overall, the study achieved its main objectives and demonstrated the potential of combining sensors, camera technology, and mobile communication to improve home security.

Recommendations

Based on the results and findings of the study, the following recommendations are proposed:

1. **System Enhancement:** Future developers are encouraged to improve the system by adding features such as sound alarms, night vision cameras, or facial recognition to make detection more accurate and useful during both day and night.

2. **Cloud Storage and Backup:** Integrating cloud storage for captured images and videos is recommended to ensure that

important data is saved even if the device is damaged or tampered with.

3. **Wider Coverage:** The system can be expanded to include multiple sensors and cameras for larger homes or multi-room coverage.

4. **Power Backup Integration:** A rechargeable battery or solar panel system can be added to keep the system running even during power outages.

5. **User Training and Awareness:** Homeowners should be oriented on how to properly use and maintain the system to ensure its efficiency and longevity.

6. **Further Research:** It is recommended that future studies explore the use of Artificial Intelligence (AI) to reduce false alarms and improve detection accuracy. Additional studies may also test the system in different environments such as offices, schools, or small businesses.

7. **Community Application:** The system can also be promoted for barangay or neighborhood use, creating a network of connected sensors that can notify nearby residents during potential security threats

Final Findings

In conclusion, the Intrusion Detection System using Motion Sensor is a promising innovation that strengthens home safety through affordable and modern technology. It serves as a step towards smarter, safer, and more connected communities, reflecting the importance of technology in improving everyday security.